Interpreting Cumulative Frequency Diagrams

Exam code: 0607

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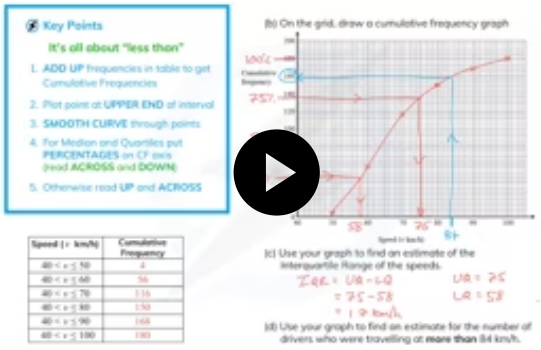
Updated on 21 January 2026



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Interpreting cumulative frequency diagrams

How do I use and interpret a cumulative frequency diagram?

- A cumulative frequency diagram provides a way to **estimate** key facts about the data
  - **median**
  - **lower** and **upper quartiles** (and **interquartile range**)
  - **percentiles**
- These values will be **estimates** as the original raw data is unknown
  - Cumulative frequency diagrams are used with **grouped** data
  - Points are joined by a smooth curve
    - This means the data is assumed to be **smoothly spread out** over each interval

How do I find the median, lower quartile and upper quartile from a cumulative frequency diagram?

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- If not, it will be the **highest value** on the frequency (y-) axis that the curve on the diagram reaches
  - This should be "top right" of the curve on a cumulative frequency diagram

- Finding the **median**:
  - **STEP 1**
    Find the **position** of the median, for n data values, this will be $\frac{n}{2}$
    - This is different from finding the median from a set of data values
    - E.g. for a list of 60 data values the median would be halfway between the 30th and 31st values
    - But for a cumulative frequency diagram it would just be $\frac{60}{2} = 30$
  - **STEP 2**
    Draw a **horizontal line** from $\frac{n}{2}$ on the cumulative frequency (y-) axis until it hits the curve
  - **STEP 3**
    Draw a **vertical line** from that point on the curve down to the horizontal (x-) axis
    - The value where that line hits the horizontal axis will be the median

- Finding the **lower quartile**:
  - **STEP 1**
    Find the **position** of the lower quartile
    - For n data values this will be $\frac{n}{4}$
  - **STEP 2**
    Draw a **horizontal line** from $\frac{n}{4}$ on the cumulative frequency axis until it hits the curve
  - **STEP 3**
    Draw a **vertical line** from that point on the curve down to the horizontal (x-) axis
    - The value where that line hits the horizontal axis will be the lower quartile

- Finding the **upper quartile**:
  - **STEP 1**
    Find the **position** of the upper quartile
    - For n data values this will be $\frac{3n}{4}$ $\left(\text{i.e. } 3 \times \frac{n}{4}\right)$
  - **STEP 2**
    Draw a **horizontal line** from $\frac{3n}{4}$ on the cumulative frequency axis until it hits the curve
  - **STEP 3**
    Draw a **vertical line** from that point on the curve to the horizontal (x-) axis
    - The value where that line hits the horizontal axis will be the upper quartile

How do I find a percentile from a cumulative frequency diagram?

- The **25th** and **75th percentiles** are the same as the **lower** and **upper quartiles** (respectively)

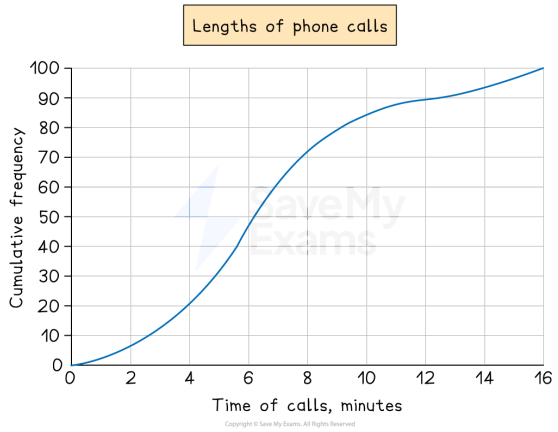
- To find the **p^{th} percentile**:
  - **STEP 1**
    Find the **position** of the pth percentile
    - For n data values, this will be $\frac{np}{100}$ $\left(\text{i.e. } \frac{n}{100} \times p\right)$
    - So for the 10th percentile ($p = 10$) with 60 data values ($n = 60$)
    - The position is $\frac{10}{100} \times 60 = \frac{1}{10} \times 60 = 6$
  - **STEP 2**
    Draw a **horizontal line** from $\frac{np}{100}$ on the cumulative frequency axis until it hits the curve
  - **STEP 3**
    Draw a **vertical line** from that point on the curve down to the horizontal (x-) axis
    - The value where that line hits the horizontal axis will be the pth percentile

Worked Example

A company is investigating the length of telephone calls customers make to its help centre.

The company randomly selects 100 phone calls from a particular day.

The results are displayed in the cumulative frequency diagram below.



(a) Estimate the median, the lower quartile and the upper quartile.

Answer:

There are 100 pieces of data, so $n = 100$

$$\frac{n}{4} = \frac{100}{4} = 25$$

$$\frac{3n}{4} = 3 \times 25 = 75$$

So the median is the 50th value
The lower quartile is the 25th value
The upper quartile is the 75th value

Draw horizontal lines from these on the cumulative frequency axis until they hit the curve
Then draw vertical lines down to the time of calls axis and take readings

Median = 6.2 minutes (6 m 12 s)
Lower quartile = 4.2 minutes (4 m 12 s)
Upper quartile = 8.2 minutes (8 m 12 s)

There is no need to convert to minutes and seconds unless the question asks you to
However, writing 6 m 2 s or 6 m 20 s would be incorrect

(b) The company is thinking of putting an upper limit of 12 minutes on calls to its help centre.

Estimate the number of these 100 calls that would have been beyond this limit.

Answer:

Draw a vertical line up from 12 minutes on the time of calls axis until it hits the curve
Then draw a horizontal line across to the cumulative frequency axis and take a reading (in this case, 90)

This tells us that up to 12 minutes, 90 of the calls had been accounted for

The question wants the number of calls that were **greater** than 12 minutes so subtract this from the total of 100

$$100 - 90 = 10$$

Approximately 10 (out of 100) calls were beyond the 12 minute limit

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